Pro Football Sports Science Research Update

September 2023

Accuracy of a smart bottle in measuring fluid intake by American football players during pre-season training. Scientific Reports, 12:11383, 2023.

Assessing isometric kicking force and post-match responses using the Kicker test. Journal of Sports Sciences, Ahead of Print: 1-8, 2023.

Association between head impact exposure, psychological needs, and indicators of mental health among U.S. high school tackle football players. Journal of Adolescent Health, Ahead of Print: 1-8, 2023.

Association of injury rates among players in the National Football League with playoff qualification, travel distance, game timing, and the addition of another game – Data from the 2017 to 2022 seasons. The Orthopaedic Journal of Sports Medicine, 11(8): 1-7, 2023.

Cannabis and sport: A World Anti-Doping perspective. Addiction, Ahead of Print: 1-3, 2023.

Cannabis use for exercise recovery in trained individuals: A survey study. Journal of Cannabis Research, 5:32: 1-8, 2023.

Core body temperatures in intermittent sports: A systematic review. Sports Medicine, Ahead of Print: 1-24, 2023.

Food-derived bioactive peptides as momentous food components: Can functional peptides passed through the PI3K/Akt/mTOR pathway and NF-kB pathway to repair and protect the skeletal muscle injury? Critical Reviews in Food Science and Nutrition, Ahead of Print: 1-19, 2023.

Hamstring injuries in the National Football League: An epidemiological study. Journal of Orthopaedics, 35: 111-114, 2023.

Hamstring muscle injury is preceded by a short period of higher running demands in professional football players. Biology of Sport, 41(1): 227-233, 2024.

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Pro Football Sports Science Research Update

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Intraindividual variability in sleep among athletes: A systematic review of definitions, operationalizations, and key correlates. Scandinavian Journal of Medicine and Science in Sports, Ahead of Print: 1-10, 2023.

Low carbohydrate availability impairs hypertrophy and anaerobic performance. Current Opinion in Clinical Nutrition and Metabolic Care, 26: 347-352, 2023.

Mechanisms of mechanical overload-induced skeletal muscle hypertrophy: Current understanding and future directions.

Mental fatigue impairs tackling technique in amateur rugby union players. International Journal of Sports Physiology and Performance, Ahead of Print: 1-8, 2023.

The effect of pre-exercise hyperhydration on exercise performance, physiological outcomes and gastrointestinal symptoms: A systematic review. Sports Medicine, Ahead of Print: 1-24, 2023.

The effect of Thursday night football on injuries in the National Football League. Journal of Orthopaedics and Sports Medicine, 5: 304-310, 2023.

Thinking about elite performance: The experience and impact of mental fatigue in elite sport coaching. International Journal of Sports Physiology and Performance, Ahead of Print: 1-7, 2023.

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Pro Football Sports Science Research Update

Accuracy of a smart bottle in measuring fluid intake by American football players during pre-season training. Scientific Reports, 12:11383, 2023.

Abstract - Rehydration is important for athlete performance and recovery. However, it can be challenging to follow appropriate fluid replacement practices due in part to difficulties in tracking fluid intake in real time. The purpose of this study was to determine the accuracy of a smart bottle in measuring fluid intake during exercise. Thirty male American football athletes drank from bottles equipped with a smart cap during outdoor pre-season practices (110± 30 min; 29.3 ± 3.0 °C; 75 ± 11% rh). The cap technology included optical sensors, microprocessors, batteries, and wireless connectivity that transmitted fluid volume data to a smartphone application in real-time. Reference measurements of fluid intake from the smart bottle were determined by gravimetry followed by conversion to volume using the density of the fluid consumed. There was no significant mean difference in fluid intake between the smart bottle and reference method (1220 ± 371 ml vs. 1236 ± 389 ml, p= 0.39 paired t test). Bland–Altman 95% limits of agreement between methods was − 212 to 180 ml. The smart bottle provided accurate measurements of fluid intake during exercise in real-world field conditions on a group level and within limits of agreement of − 212 to 180 ml (or approximately ± 15% of overall fluid intake) on an individual level.

Assessing isometric kicking force and post-match responses using the Kicker test. Journal of Sports Sciences, Ahead of Print: 1-8, 2023.

Abstract - This study examined the inter-session reliability of force output from a novel isometric strength assessment protocol (the Kicker); and its suitability to monitor soccer player's combined hip flexion and knee extension force capacity over 72-h post-competitive matches. Reliability (Part-A) testing was completed over three sessions on 20 individuals participating in various sports at a recreational level or higher. Post-match strength response (Part-B) data were collected for 72-h after a game (24-h (+24), 48-h (+48) and 72- h (+72) post-match) in 17 male academy soccer players. After familiarization, Kicker force for each limb showed high inter-session reliability (ICC >0.95; typical error <14 N, CV <6%); minimum detectable change at a 95% confidence interval <40 N). Across the 72-h post-match period, Kicker force for each limb was suppressed compared to baseline (force loss range = −5.8% to −12.5%; effect sizes range = −0.26 to −0.43) at all time points. The Kicker assessment protocol measures combined isometric hip flexor and knee extensor force capacity with high inter-session

reliability. The proof of concept that the protocol can be used as a monitoring tool was evidenced by sustained suppression of baseline force capacity in both kicking limbs for 72-h post soccer matches.

Association between head impact exposure, psychological needs, and indicators of mental health among U.S. high school tackle football players. Journal of Adolescent Health, Ahead of Print: 1-8, 2023.

Purpose: Age of first exposure to tackle football and head impact kinematics have been used to examine the effect of head impacts on mental health outcomes. These measures coupled with retrospective and cross-sectional designs have contributed to conflicting results. The purpose of this study was to identify the effect of one season of head impact exposure, age of first exposure to football, and psychological need satisfaction on acute mental health outcomes in adolescent football players.

Methods: This prospective single-season cohort study used sensor-installed mouthguards to collect head impact exposure along with surveys to assess age of first exposure to football, psychological satisfaction, depressive symptoms, anxiety symptoms, and thriving from football players at four high schools (n = 91). Linear regression was used to test the association of head impact exposure, age of first exposure, and psychological satisfaction with acute mental health outcomes.

Results: A total of 9,428 impacts were recorded with a mean of 102 ± 113 impacts/player. Cumulative head impact exposure and age of first exposure were not associated with acute mental health outcomes at postseason or change scores from preseason to postseason. Greater psychological satisfaction was associated with fewer depressive symptoms (β = -0.035, SE = 0.008, p = < .001), fewer anxiety symptoms (β = -0.021, SE = 0.008, p = .010), and greater thriving scores (β = 0.278, SE = 0.040, p = < .001) at postseason.

Discussion: This study does not support the premise that greater single-season head impact exposure or earlier age of first exposure to tackle football is associated with worse acute mental health indicators over the course of a single season in adolescent football players.

Pro Football Sports Science Research Update

Association of injury rates among players in the National Football League with playoff qualification, travel distance, game timing, and the addition of another game – Data from the 2017 to 2022 seasons. The Orthopaedic Journal of Sports Medicine, 11(8): 1-7, 2023.

Background: Injury incidence is higher in the National Football League (NFL) than in other collision sports. Although previous research has identified that scheduling variations, including overseas games and bye week timing, does not affect concussion risk, data are currently lacking regarding the effects of scheduling variation on season-long musculoskeletal injury incidence.

Purpose: To determine whether higher cumulative travel distance, overseas play, an early season bye week, and an expansion of the regular season is associated with higher injury rates in the NFL.

Study Design: Descriptive epidemiology study.

Methods: All 1275 injuries across 5 NFL seasons from 2017 to 2018 through 2021 to 2022 were reviewed retrospectively. Injury data and travel distances were extracted from publicly available sources, which were cross-referenced for validation. Injury rates were calculated per 1000 athletic-exposures (AEs). Cumulative team travel distances were compared statistically using a linear regression. Single factor analysis of variance was used to compare categorical variables.

Results: Travel distance did not significantly predict injury rates (P = .47), and there was no difference in injury rates between teams that played a game overseas versus teams that did not (19.3 injuries per 1000 AEs for both; P = .96). In addition, no difference was found in injury rates (F[109,2100[= 0.704; P = .73) or players placed on the injured reserve list (F[99,778] = 1.70; P = .077) between various bye weeks (P = .73). Injury rates did not differ between the new 17-game regular season (18.4 per 1000 AEs) versus the previous four 16-game regular seasons (19.7 per 1000 AEs; P = .12). However, teams that did not qualify for the playoffs had a significantly higher injury rate (19.9 per 1000 AEs) as well as players on injured reserve (8.0 per 1000 AEs) than playoff-qualifying teams (18.4 and 6.8 per 1000 AEs, respectively; P < .05 for both).

Conclusion: Over 5 NFL seasons, cumulative travel distance, overseas play, bye week timing, and adding 1 regular season game were not associated with increased injury rates in NFL players. However, a lack of regular season success was associated with higher injury rates and more players on injured reserve.

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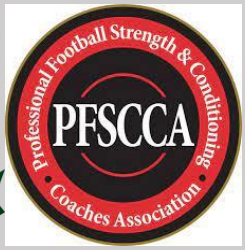


Pro Football Sports Science Research Update

Cannabis and sport: A World Anti-Doping perspective. Addiction, Ahead of Print: 1-3, 2023.

"It is the opinion of the World Anti-Doping Agency (WADA) that cannabis should remain a banned substance in sport. The purpose of this editorial is to briefly summarize the elements of the discussion leading to the continued ban in sport."

Cannabis availability and its subsequent use, misuse and incidence of use disorder are increasing globally [1], even as the potency of the plant, gauged by delta-9-tetrahydrocannabinol (Δ9-THC content), continues to increase [2]. It is not known what proportion of top-level athletes regularly consume cannabis. However, a recent review of self-reports that included more than 46000 athletes suggests that the proportion of athletes using cannabis is no different from the age-matched general population [3]. The World Anti-Doping Agency (WADA) banned cannabis in competition in all sports when the 2004 List of Prohibited Substances and Methods was first released under the World Anti-Doping Code [4]. It has since revisited and updated the cannabinoid category to include synthetic cannabinoids, increase reporting thresholds and to specifically exempt cannabidiol. In response to stakeholder requests, the WADA Executive Committee asked for a review of the status of cannabis. Ad hoc committees conducted a scientific, ethical and medical review of the effects of cannabis, focusing upon the primary psychoactive ingredient, Δ9-THC, among the more than 100 distinct cannabinoids within the cannabis plant.

For any substance or method to be considered for inclusion in the Prohibited Substances List as per the World Anti-Doping Code, two of the following three equally-important criteria must be met: (1) it enhances or has the potential to enhance sport performance, (2) it represents an actual or potential risk to the health of the athlete and (3) it violates the spirit of the sport as defined in the Code.

In 2022, WADA reviewed more than 2700 articles for the present exercise. Inclusion of a scientific study or review to be further discussed was aided by adherence to scientific principles, where appropriate. This included application of rigorous methods, such as randomization of subjects to treatment groups, use of placebo controls and blinding of both subject and experimenter. Publication in a peer-reviewed scientific journal was required, and greater emphasis was placed upon studies that had results which were independently replicated. This enabled the Expert List, Athletic, Ethical and Health, Medical and Research committees to review the most current science and issue independent opinions on the

subject. Finally, as part of the scientific review process, world leading experts on cannabis and addiction behavior were consulted to validate the conclusions of the committees.

Perspectives were mixed on the first List inclusion criterion. There were no rigorously conducted studies reviewed that indicated any direct ergogenic effect of cannabis. Nonetheless, a number of reports suggested that other benefits may be conferred by use of the drug, such as reduction in pain or anxiety [5, 6] associated with athletic competition. Some athletes report benefit following athletic performance, by facilitating recovery and reducing pain. Hence, while there may be benefits perceived by some athletes, there were no scientific data to support performance enhancement.

There was far more agreement on the second criterion—actual or potential risk to the health of the athlete. There is a comprehensive historical literature as well as a rapidly growing body of contemporary literature supporting the assertion that cannabis use can negatively impact the health, safety or wellbeing of the athlete. Acute intoxication can result in deficits in reaction time, temporal estimation and dexterity [7-11] as well as in psychiatric symptoms [12]. Not all studies detected similar changes; results will vary with the dose, route, subjects' experience and test methods employed. Chronic, frequent use of cannabis is associated with an increased risk of cannabis use disorder, as recently reviewed by Robinson and colleagues [13], as well as potentially other long-term changes [14, 15], including an increased risk of precipitation of severe mental illness [16]. There is a growing body of evidence that cannabis use can affect brain development and anatomy in adolescents and young adults [17-19], although the consequences of these changes are not yet firmly established. The potential of cannabis use to compromise the health and safety of the athlete qualifies the application of this criterion.

The third criterion, spirit of the sport, is considered as important as the other two criteria. The Ethics Advisory Group noted that spirit of sport encompasses a number of universal values of sport, and four aspects that were particularly relevant to the discussion on cannabis remaining in the list included the following:

- Excellence in performance: this could be undermined by consumption of cannabis during the in-competition period.
- Character and education: the role model aspect is not compatible with use of a substance that is still illegal in most parts of the world.
- Respect for rules and laws: its use violates the law in most countries in the world, in addition to Anti-Doping Organization rules in some instances.

- Respect for self and other participants: the welfare and safety of other participants may be compromised by impaired judgment associated with the presence of cannabis in an athlete in-competition.

Based upon all the above considerations, which included input from the Ethics and Athletic committees at WADA, it was concluded that cannabis use met the Spirit of Sport criterion.

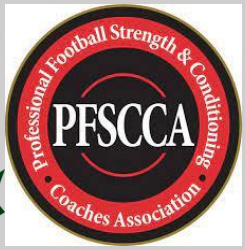
WADA emphasizes that prohibition of cannabis is in-competition only, which is defined as after 23:59 hours on the day prior to competition. The current decision limit of 180 ng/ml of Δ9-THC-COOH in urine and a cut-off of 150 ng/ml, plus the uncertainty of measurement of 30 ng/ml, takes this into account. Because of these high thresholds, primarily chronic, frequent cannabis users and athletes consuming high doses in-competition will be detected. Therefore, the cut-off generally will not affect the freedom of an athlete who wishes to legally consume cannabis outside of competition. Athletes who have a need for medicinal cannabis treatment should request a therapeutic use exemption (TUE).

Cannabis use for exercise recovery in trained individuals: A survey study. Journal of Cannabis Research, 5:32: 1-8, 2023.

Background: Cannabis use, be it either cannabidiol (CBD) use and/or delta-9-tetrahydrocannabinol (THC) use, shows promise to enhance exercise recovery. The present study aimed to determine if individuals are using CBD and/or THC as a means of recovery from aerobic and/or resistance exercise, as well as additional modalities that might be used to aid in recovery.

Methods: Following consent, 111 participants (Mean ± SD: Age: 31±13 years) completed an anonymous survey. All participants were regularly using cannabis (CBD and/or THC) as well as were currently exercising. Questions pertained to level of cannabis use, methods used for consumption of cannabis, exercise habits, exercise recovery strategies, and demographics.

Results: Eighty-five percent of participants reported participating in aerobic training. In addition, 85% of participants also reported regular participation in resistance exercise. Seventy-two percent of participants participated in both aerobic and resistance exercise. Ninety-three percent of participants felt that CBD use assisted them with recovery from exercise, while 87% of participants felt the same regarding THC use.



Pro Football Sports Science Research Update

Conclusions: Individuals who habitually use cannabis, CBD or THC, and regularly engage in exercise do feel that cannabis assists them with exercise recovery. More data are necessary to understand the role of cannabis in exercise recovery as well as perceived ergogenic benefits of cannabis by individuals who both regularly participate in exercise and habitually use cannabis

Core body temperatures in intermittent sports: A systematic review. Sports Medicine, Ahead of Print: 1-24, 2023.

Background: Hyperthermia (and associated health and performance implications) can be a significant problem for athletes and teams involved in intermittent sports. Quantifying the highest thermal strain (i.e. peak core body temperature [peak Tc]) from a range of intermittent sports would enhance our understanding of the thermal requirements of sport and assist in making informed decisions about training or match-day interventions to reduce thermally induced harm and/or performance decline.

Objective: The objective of this systematic review was to synthesize and characterize the available thermal strain data collected in competition from intermittent sport athletes.

Methods: A systematic literature search was performed on Web of Science, MEDLINE, and SPORTDiscus to identify studies up to 17 April 2023. Electronic databases were searched using a text mining method to provide a partially automated and systematic search strategy retrieving terms related to core body temperature measurement and intermittent sport. Records were eligible if they included core body temperature measurement during competition, without experimental intervention that may influence thermal strain (e.g. cooling), in healthy, adult, intermittent sport athletes at any level. Due to the lack of an available tool that specifically includes potential sources of bias for physiological responses in descriptive studies, a methodological evaluation checklist was developed and used to document important methodological considerations. Data were not meta-analyzed given the methodological heterogeneity between studies and therefore were presented descriptively in tabular and graphical format.

Results: A total of 34 studies were selected for review; 27 were observational, 5 were experimental (2 parallel group and 3 repeated measures randomized controlled trials), and 2 were quasi-experimental (1 parallel group and 1 repeated measures non-randomized controlled trial). Across all included studies, 386 participants (plus participant numbers not reported in two studies) were recruited after accounting

for shared data between studies. A total of 4 studies (~12%) found no evidence of hyperthermia, 24 (~71%) found evidence of 'modest' hyperthermia (peak Tc between 38.5 and 39.5 °C), and 6 (~18%) found evidence of 'marked' hyperthermia (peak Tc of 39.5 °C or greater) during intermittent sports competition.

Conclusions: Practitioners and coaches supporting intermittent sport athletes are justified to seek interventions aimed at mitigating the high heat strain observed in competition. More research is required to determine the most effective interventions for this population that are practically viable in intermittent sports settings (often constrained by many competing demands). Greater statistical power and homogeneity among studies are required to quantify the independent effects of wet bulb globe temperature, competition duration, sport and level of competition on peak Tc, all of which are likely to be key modulators of the thermal strain experienced by competing athletes.

Key Points

- Intermittent sport athletes generate core body temperatures in competition that surpass magnitudes associated with impaired health and performance.
- Practitioners and coaches supporting intermittent sport athletes are justified to seek interventions aimed at mitigating the high heat strain observed in competition.
- More research is required to determine the independent effects of wet bulb globe temperature, competition duration, sport and level of competition on peak core body temperature—all are likely to be key modulators of the thermal strain experienced by competing athletes.

Food-derived bioactive peptides as momentous food components: Can functional peptides passed through the PI3K/Akt/mTOR pathway and NF-kB pathway to repair and protect the skeletal muscle injury? Critical Reviews in Food Science and Nutrition, Ahead of Print: 1-19, 2023.

Abstract - Muscle injury is defined as an overuse injury or traumatic distraction of a muscle, which is latent in any sport event, from amateur to large events. Based on previous numbers of muscle injuries and time spent to the athletes' recovery, the use of dietary functional factors intervention strategies is essential to enhance the recovery process and health. In recent years, there has been increasing

evidence that biologically active peptides played an important role in sports nutrition and muscle injure recovery. Food-derived bioactive peptides were physiologically active peptides mostly derived from proteins following hydrolysis, which could be resorbed in intact form to reduce muscle damage following exercise and induce beneficial adaptions within the connective tissue. However, the complexity of the histoarchitectural considerations for skeletal muscle injuries and the repair mechanism of damaged skeletal muscle were not well known. In the following overview, the potential mechanisms and possible limitations regarding the damaged skeletal muscle metabolism were summarized, which aimed to present an overview of the nutritional strategies and recommendations after a muscular sports injury, emphasizing the use of main bioactive peptides. In addition, this review will provide implications for the studies of dietary bioactive peptides in the future.

Hamstring injuries in the National Football League: An epidemiological study. Journal of Orthopaedics, 35: 111-114, 2023.

Background: While numerous studies have evaluated National Football League injuries, there is limited literature evaluating hamstring injuries sustained in games. Our primary aim is to analyze the effect of player position on the relative incidence of hamstring injuries in the National Football League. Our secondary aims are to analyze the effects of field surface, week of the season, and short rest weeks.

Methods: Official National Football League game books containing injury data from the 2013–2016 regular seasons were used. Data were analyzed to determine the incidence of hamstring injuries by field surface, rest, and week of the season. Field surface was considered either turf or grass. Short rest was considered four days. Relative incidence of hamstring injuries by position was performed with standardized incidence ratios. P values < 0.05 were considered statistically significant.

Results: Seventy-eight qualifying hamstring injuries were identified and included in our analysis. Linebackers had the highest relative incidence per play with a standardized incidence ratio of 2.02 (CI: 1.14–2.91), followed by Defensive Backs (1.62; 95% CI: 1.14–1.62). Offensive linemen and defensive linemen had standardized incidence ratios significantly less than 1. Fifty-seven percent of hamstring injuries occurred on turf fields (p = 0.082). There was no significant difference between the proportion of hamstring injuries that occurred on short rest and the proportion of games played on short rest (p = 0.959). Hamstring injuries were not more likely to occur than the pooled group of all other types of

injuries on short rest (p = 0.861). With a 17-week season, the mean week of hamstring injury was 8.05 (95% CI: 7.06–9.04), while the median week was 7.5.

Conclusions: Linebackers and Defensive Backs have the highest relative incidence of hamstring injuries compared to other position groups, while offensive and defensive linemen have the lowest. Field surface and a short rest period did not show significance.

Hamstring muscle injury is preceded by a short period of higher running demands in professional football players. Biology of Sport, 41(1): 227-233, 2024.

Abstract - The aim of this study was to examine match running patterns before a hamstring muscle injury occurs during a match in male professional football players. A total of 281 male professional football players belonging to 7 teams from LaLiga were prospectively monitored over three seasons. Among these, 36 players suffered a non-contact hamstring muscle injury during an official match. The injuries were recorded by the medical staff, including the minute when the injury occurred. Running distances at different speed thresholds for 5 min and 15 min before the injury were compared to mean values of the previous 5 matches for the same time points. There were a total of 44 non-contact hamstring muscle injuries, which represents a hamstring muscle injury incidence of 3.34 injuries/1000 h of match exposure. The average time loss for these injuries was 33±28 days (range 7 to 117 days). In the 15 min prior to the injury, players ran a similar distance as in control matches (p from 0.22 to 0.08). However, players ran a greater distance in the 5-min period before the injury than in control matches at 21.0–23.9 km/h ($p < 0.001$) and at ≥ 24 km/h ($p < 0.001$). The odds ratio for a hamstring muscle injury was 7.147 for those players who ran > 30.0 m at ≥ 21 km/h in a 5-min period ($p < 0.001$). Hamstring muscle injuries during competition were preceded by 5 min of higher running demands at > 21 km/h, compared with control matches. This suggests that a short period of unusual running increases the risk of hamstring muscle injury in professional football players.

Pro Football Sports Science Research Update

Impact of lumbar disk herniation on performance outcomes and new contracts in the National Football League. Clinical Spine Surgery, 36(4): E139-E144, 2023.

Study design: This was a retrospective cohort study.

Objective: To determine performance outcomes and the contract-signing ability for the most recent cohort of professional football players treated for lumbar disk herniation (LDH).

Summary of background data: LDH can have a significant impact on the career of a National Football League (NFL) player. Previous studies have found favorable return to play (RTP) and performance outcomes for players with LDH, but the impact on the ability to sign new contracts (an important surrogate to assess continued success) has not previously been studied.

Materials and methods: NFL players treated for LDH from 2000 to 2020 were identified from a public records search. Age, position, type of treatment, and RTP measures were collected. Pro Football Focus (PFF) performance grade and contract values were compared before the injury and after treatment. Multivariable logistic regression was used to identify independent risk factors associated with the ability to RTP and sign high-value contracts.

Results: One hundred one players were treated for an LDH, of which 75 returned to play. Posttreatment performance as measured by PFF was similar to preinjury levels ($P = 0.2$). However, both total and guaranteed contract values were significantly reduced ($P < 0.01$). In multivariable analysis, both lower age and higher preinjury PFF grade were independent predictors of RTP and ability to sign a new contract. A preinjury contract that contained a high proportion of guaranteed money was found to be an independent predictor of the ability to sign a contract that was >20% guaranteed.

Conclusion: Although the majority of players were able to RTP at preserved performance levels following LDH treatment, their contract values were significantly reduced. RTP and contract-signing ability were not associated with the type of treatment, but rather baseline factors such as the player's age, performance, and preinjury compensation.

Level of evidence: Level IV.

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Pro Football Sports Science Research Update

Intraindividual variability in sleep among athletes: A systematic review of definitions, operationalizations, and key correlates. Scandinavian Journal of Medicine and Science in Sports, Ahead of Print: 1-10, 2023.

Abstract - Via systematic review with narrative synthesis of findings, we aimed to document the ways by which researchers have defined, operationalized, and examined sleep variability among athletes. We identified studies in which scholars examined in-traperson variability in sleep among athletes via a search of six databases (Web of Science, Embase, Medline, PsycINFO, CINHAL Plus, and ProQuest Dissertations and Theses Global) using a protocol that included keywords for the target outcome (sleep*), population (athlet* OR sport*), and outcome operationalization (variability OR variation OR "standard deviation" OR fluctuate OR fluctuation OR stability OR instability OR reactivity OR IIV OR intraindividual). We complemented this primary search with citation searching of eligible articles. Assessments of study quality captured eight core elements, namely aims/hypotheses, sample size justification, sample representativeness, number of days sleep assessed, measures of sleep and its correlates, missing data, and inferences and conclusions. From a total of 1209 potentially relevant papers, we identified 16 studies as meeting our eligibility criteria. Concept definitions of variability were notably absent from this work and where available were vague. Quantitative deviations from one's typical level of target sleep metrics reflected the essence by which all but one of the research teams operationalized sleep variability. We assessed the overall quality of empirical work as moderate in nature. We propose a working definition of sleep variability that can inform knowledge generation on the temporal, day-to-day dynamics of sleep functioning that is required for personalized interventions for optimizing sleep health.

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Pro Football Sports Science Research Update

Low carbohydrate availability impairs hypertrophy and anaerobic performance. Current Opinion in Clinical Nutrition and Metabolic Care, 26: 347-352, 2023.

Purpose: of review Highlight contemporary evidence examining the effects of carbohydrate restriction on the intracellular regulation of muscle mass and anaerobic performance.

Recent findings: Low carbohydrate diets increase fat oxidation and decrease fat mass. Emerging evidence suggests that dietary carbohydrate restriction increases protein oxidation, thereby limiting essential amino acid availability necessary to stimulate optimal muscle protein synthesis and promote muscle recovery. Low carbohydrate feeding for 24 h increases branched-chain amino acid (BCAA) oxidation and reduces myogenic regulator factor transcription compared to mixed-macronutrient feeding. When carbohydrate restriction is maintained for 8 to 12 weeks, the alterations in anabolic signaling, protein synthesis, and myogenesis likely contribute to limited hypertrophic responses to resistance training. The blunted hypertrophic response to resistance training when carbohydrate availability is low does not affect muscle strength, whereas persistently low muscle glycogen does impair anaerobic output during high-intensity sprint and time to exhaustion tests.

Summary: Dietary carbohydrate restriction increases BCAA oxidation and impairs muscle hypertrophy and anaerobic performance, suggesting athletes who need to perform high-intensity exercise should consider avoiding dietary strategies that restrict carbohydrate.

Key Points

- Low carbohydrate diets decrease body fat mass and increase whole-body and skeletal muscle fat oxidation.
- New data demonstrates that low carbohydrate diets increase the concentrations of BCAA and muscle protein breakdown metabolites, due to increased reliance on essential amino acids for oxidation.
- Increased amino acid oxidation in response to low carbohydrate availability decreases muscle remodeling and repair after exercise, likely contributing to blunted hypertrophic responses to resistance exercise training.
- Impaired muscle hypertrophy does not affect muscle strength, but persistently low glycogen following a low carbohydrate diet diminishes anaerobic performance

Pro Football Sports Science Research Update

Mechanisms of mechanical overload-induced skeletal muscle hypertrophy: Current understanding and future directions. Physiological Reviews, Ahead of Print: 1-126, 2023.

Abstract - Mechanisms underlying mechanical overload-induced skeletal muscle hypertrophy have been extensively researched since the landmark report by Morpurgo (1897) of "work-induced hypertrophy" in dogs that were treadmill-trained. Much of the pre-clinical rodent and human resistance training research to date supports that involved mechanisms include enhanced mammalian/mechanistic target of rapamycin complex 1 (mTORC1) signaling, an expansion in translational capacity through ribosome biogenesis, increased satellite cell abundance and myonuclear accretion, and post-exercise elevations in muscle protein synthesis rates. However, several lines of past and emerging evidence suggest additional mechanisms that feed into or are independent of these processes are also involved. This review will first provide a historical account as to how mechanistic research into skeletal muscle hypertrophy has progressed. A comprehensive list of mechanisms associated with skeletal muscle hypertrophy is then outlined and areas of disagreement involving these mechanisms are presented. Finally, future research directions involving many of the discussed mechanisms will be proposed.

Mental fatigue impairs tackling technique in amateur rugby union players. International Journal of Sports Physiology and Performance, Ahead of Print: 1-8, 2023.

Purpose: To test the effects of mental fatigue (MF) on tackling technique on the dominant and nondominant shoulders in rugby union.

Methods: Twenty male amateur rugby union players and a total of 953 tackles were analyzed. A randomized crossover counterbalanced design was used across a non-MF (control) and an MF condition. During each condition, each player performed 24 tackles, divided into 4 sets of 6 tackles (3 tackles on each shoulder). In the MF condition, players performed the Stroop Task between each set of tackles. A video recording of each tackle was used to evaluate each player's technical proficiency. A score of 1 point was awarded if a specific technique was performed correctly, and 0 point was given if not. The total score, measured in arbitrary units (AU) out of 11, represents the player's overall tackling proficiency.

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Results: Overall, players displayed a significantly lower technical proficiency score in the MF condition compared to control (set 2: control 7.30 [7.04–7.57] AU vs MF 6.91 [6.70–7.12] AU, P = .009, effect size (ES) = 0.30 small and set 3: control 7.34 [7.11–7.57] AU vs MF 6.88 [6.66–7.11] AU, P = .002, ES = 0.37 small). For the nondominant shoulder, players had a significantly lower technical proficiency score during the MF condition at set 2 (control 7.05 [6.68–7.41] AU vs MF 6.69 [6.42–6.96] AU, P = .047, ES = 0.29 small) and set 3 (control 7.14 [6.83–7.45] AU vs MF 6.61 [6.35–6.87] AU, P = .007, ES = 0.49 small).

Conclusions: MF can diminish a player's overall tackling proficiency, especially when tackling on the nondominant shoulder. The physiological mechanism for this finding may be impaired executive function and suboptimal functioning of neural signals and pathways, which result in less skillful coordination of movement. To further understand and explain MF-induced physiological changes in tackling, the feasibility of monitoring brain activity (such as electroencephalogram) and neuromuscular function (such as electromyogram) needs to be investigated. The findings from this study may also contribute to the development of more effective tackle training programs for injury prevention and performance.

The effect of pre-exercise hyperhydration on exercise performance, physiological outcomes and gastrointestinal symptoms: A systematic review. Sports Medicine, Ahead of Print: 1-24, 2023.

Background: Fluid loss during prolonged exercise in hot conditions poses thermoregulatory and cardiovascular challenges for athletes that can lead to impaired performance. Pre-exercise hyperhydration using nutritional aids is a strategy that may prevent or delay the adverse effects of dehydration and attenuate the impact of heat stress on exercise performance.

Objectives: The aim of this systematic review was to examine the current literature to determine the effect of pre-exercise hyperhydration on performance, key physiological responses and gastrointestinal symptoms.

Methods: English language, full-text articles that compared the intervention with a baseline or placebo condition were included. An electronic search of Medline Complete, SPORTDiscus and Embase were used to identify articles with the final search conducted on 11 October 2022. Studies were assessed using the American Dietetic Association Quality Criteria Checklist.

Pro Football Sports Science Research Update

Results: Thirty-eight studies involving 403 participants (n=361 males) were included in this review (n=22 assessed exercise performance or capacity). Two studies reported an improvement in time-trial performance (range 5.7–11.4%), three studies reported an improvement in total work completed (kJ) (range 4–5%) and five studies reported an increase in exercise capacity (range 14.3–26.2%). During constant work rate exercise, nine studies observed a reduced mean heart rate (range 3–11 beats min−1), and eight studies reported a reduced mean core temperature (range 0.1–0.8 °C). Ten studies reported an increase in plasma volume (range 3.5–12.6%) compared with a control. Gastrointestinal symptoms were reported in 26 studies, with differences in severity potentially associated with factors within the ingestion protocol of each study (e.g. treatment, dose, ingestion rate).

Conclusions: Pre-exercise hyperhydration may improve exercise capacity during constant work rate exercise due to a reduced heart rate and core temperature, stemming from an acute increase in plasma volume. The combination of different osmotic aids (e.g. glycerol and sodium) may enhance fluid retention and this area should continue to be explored. Future research should utilize valid and reliable methods of assessing gastrointestinal symptoms. Furthermore, studies should investigate the effect of hyperhydration on different exercise modalities whilst implementing a strong level of blinding. Finally, females are vastly underrepresented, and this remains a key area of interest in this area.

Key Points

- Pre-exercise hyperhydration appears to acutely increase plasma volume which may reduce heart rate and core temperature during constant work rate exercise to exhaustion compared with control.
- Implementing pre-exercise hyperhydration may induce gastrointestinal symptoms during exercise in some individuals, with the severity of symptoms potentially associated with the dose, timing and ingestion rate of the hyperhydration strategy.
- The effect of hyperhydration on time-trial performance is equivocal and provides an area for future research.

Pro Football Sports Science Research Update

The effect of Thursday night football on injuries in the National Football League. Journal of Orthopaedics and Sports Medicine, 5: 304-310, 2023.

Background: This study aimed to determine the relationship of recovery period length on overall injury patterns (rates and incidence) for athletes in the National Football League.

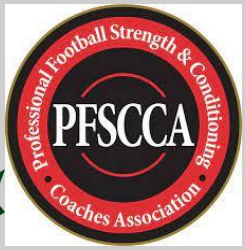
Methods: Official NFL gamebooks were queried from 2011 to 2015, and each in-game injury was analyzed for all regular season games. Data included athlete name, position, season, week, date, weekday, teams, and injured body part. ANOVA testing determined statistical significance of injury rates across weekdays, season weeks, and positions.

Results: A total of 27,712 injuries were analyzed. On average, 21.4 (8.6; 20.9-21.9) injuries were sustained per game on Sunday, 18.2 (6.9; 16.7- 19.7) on Monday, and 21.7 (6.9; 20.2-23.3) on Thursday. The difference between Sunday and Thursday games was negligible (p=0.9264), while Monday games had fewer injuries than Sundays (p=0.0028) and Thursdays (p=0.0214). There was no difference in number of injuries each team sustained on Thursday compared to the week prior to (10.8 versus 10.8; p= 0.4971) or week after (10.8 versus 10.7; p=0.7315) its Thursday game. Among positions, defensive linemen (p=0.0036), linebackers (p=0.0297), skilled running positions (p=0.0099), and wide receivers (p=0.0259) sustained fewer injuries on Mondays compared to Sundays, but no statistical difference existed between Thursdays and Sundays.

Conclusion: Our analysis indicates that shortened recovery before Thursday games does not have significant effect on injury rates compared to Sunday games, while the additional rest day before Monday games correlates to fewer injuries. Additionally, there was no residual effect of reduced rest period affecting players' risk of injury the following week. A general trend of greater injury rates occurs as the season progresses; the incidence of injury during the final week of the season is nearly double that of the first. Interestingly, although injury rates did not differ among positions between Thursdays and Sundays, defensive positions and offensive ball-carrier positions saw reduced injuries on Mondays; this effect is likely secondary to the nature of their positions being most frequently involved in high-energy collisions during tackles.

Level of Evidence: III

INFINITE ATHLETE



Pro Football Sports Science Research Update

Thinking about elite performance: The experience and impact of mental fatigue in elite sport coaching. International Journal of Sports Physiology and Performance, Ahead of Print: 1-7, 2023.

Purpose: Mental fatigue causes decreases in aspects of athletes' performance. Elite coaches commonly undertake cognitively demanding tasks and are seemingly at similar risk of subsequent performance impairment. However, elite sport coaches' experiences of mental fatigue, alongside other markers of psychobiological stress, have yet to be quantified.

Methods: Three elite coaching and performance staff (2 women and 1 man) provided 100-mm visual analog scale ratings of mental fatigue, physical fatigue, readiness to perform, and salivary samples for later cortisol (sCort) and alpha-amylase (sAA) analysis. Data were obtained on the same morning each week across a 16-week preseason. Data were subset by individual coach for descriptive and repeated-measures correlational analyses.

Results: Fluctuating mental fatigue was observed over the 16 weeks (min–max; coach 1 = 25–86 AU; coach 2 = 0–51 AU; and coach 3 = 15 − 76 AU). Elevated levels of mental fatigue were reported at multiple time points, with individual variability observed. sCort (in nanomoles per liter), sAA (in micromoles per liter), and sAA:sCort indicated that coaches experienced psychophysiological stress (min–max; coach 1 sCort = 8.42–17.31, sAA = 52.40–113.06, sAA:sCort = 3.20–12.80; coach 2 sCort = 4.20–9.70, sAA = 158.80–307.20, sAA:sCort = 21.10–61.70; and coach 3 sCort = 6.81−19.66, sAA = 86.55–495.85, sAA:sCort = 4.90–35.50). A significant inverse relationship between mental fatigue and readiness to perform ($r = -.44$ [−.64 to −.17], $P = .002$) was identified.

Conclusions: Elite sport coaches report elevated instances of mental fatigue during a preseason training period. Those involved in elite sports should act to understand the presence and potential subsequent impacts of staff mental fatigue and consider management or mitigation strategies. Optimization of the cognitive performance of coaches and performance staff presents as a potential source of competitive advantage.

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